

Fall 2024

Problem 1 (Fa24). Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function which is continuously differentiable on (a, b) . Show that for all $c \in [a, b]$, we have

$$|f(c)| \leq \frac{1}{b-a} \int_a^b |f(x)| dx + \int_a^b |f'(x)| dx$$

Problem 2 (Sp83, Sp10, Fa24). Let $\lambda_1, \lambda_2, \dots, \lambda_n, \dots$ be real numbers. Show that the infinite series

$$\sum_{n=1}^{\infty} \frac{e^{i\lambda_n x}}{n^2}$$

converges uniformly over $\mathbb{R}$ to a continuous limit function $f: \mathbb{R} \rightarrow \mathbb{C}$. Show, further, that the limit

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(x) dx$$

exists.

Problem 3 (Fa00, Fa24). Let U be a connected and simply connected open subset of the complex plane, and let f be a holomorphic function on U . Suppose a is a point of U such that the Taylor series of f at a converges on an open disc D that intersects the complement of U . Does it follow that f extends to a holomorphic function on $U \cup D$? Give a proof or a counterexample.

Problem 4 (Fa24). Let the function f be holomorphic in the strip $-1 < \Im z < 1$, real on the real axis, and of positive imaginary part in the strip $0 < \Im z < 1$.

1. Prove that f has negative imaginary part in the strip $-1 < \Im z < 0$.
2. Prove $f'(x) \geq 0$ for x real.
3. Prove $f'(x) > 0$ for x real.

Problem 5 (Sp96, Fa24). Prove or disprove: For any 2×2 matrix A over $\mathbb{C}$, there is a 2×2 matrix B such that $A = B^2$.

Problem 6 (Fa24). Let V be a finite dimensional real inner product space, and let $T: V \rightarrow V$ be a linear transformation. Assume that:

- If $\langle x, y \rangle = 0$ then $\langle T(x), T(y) \rangle = 0$ for all $x, y \in V$
- There is at least one nonzero $v \in V$ for which $\|T(v)\| = \|v\|$.

Show that T is orthogonal, i.e., $\langle T(x), T(y) \rangle = \langle x, y \rangle$ for all $x, y \in V$.

Problem 7 (Fa24). Let $\mathbb{K}$ be a field, let $\mathbb{K}[x]$ denote the polynomial ring in one variable over $\mathbb{K}$, and let $\mathcal{I}$ be an ideal in $\mathbb{K}[x]$. Show that if $\mathbb{K}[x]/\mathcal{I}$ is a field, then it is a finite extension of $\mathbb{K}$.

Problem 8 (Sp89, Fa24). Let G be a group whose order is twice an odd number. For g in G , let λ_g denote the permutation of G given by $\lambda_g(x) = gx$ for $x \in G$.

1. Let g be in G . Prove that the permutation λ_g is even if and only if the order of g is odd.
2. Let $N = \{g \in G \mid \text{ord}(g) \text{ is odd}\}$. Prove that N is a normal subgroup of G of index 2.

Problem 9 (Fa15, Fa24). Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a function, and assume that:

f is continuous on $[0, \infty)$;

f is differentiable on $(0, \infty)$;

$f'(x) \leq 0$ for all $x > 0$ such that $f(x) > 1$; and

$f(0) = 1$.

Prove that $f(x) \leq 1$ for all $x \geq 0$.

Problem 10 (Fa86, Fa24). Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by:

$$f(x, y) = \begin{cases} x^{4/3} \sin(y/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Determine all points at which f is differentiable.

Problem 11 (Su80, Fa17, Fa21, Fa24). Let f be a meromorphic function on $\mathbb{C}$ which is analytic in a neighborhood of 0. Let its Maclaurin series be

$$\sum_{k=0}^{\infty} a_k z^k$$

with all $a_k \geq 0$. Suppose there is a pole of modulus $r > 0$ and no pole has modulus $< r$. Prove there is a pole at $z = r$.

Problem 12 (Su81, Fa24). Compute

$$\frac{1}{2\pi i} \int_C \frac{dz}{\sin 1/z}$$

where C is the circle $|z| = \frac{1}{5}$ positively oriented.

Problem 13 (Sp95, Fa24). Let V be a finite dimensional vector space over a field $\mathbb{F}$, and let $L : V \rightarrow V$ be a linear transformation. Suppose that the characteristic polynomial χ of L is written as $\chi = \chi_1 \chi_2$, where χ_1 and χ_2 are two relatively prime polynomials with coefficients in $\mathbb{F}$. Show that V can be written as the direct sum of two subspaces V_1 and V_2 with the property that $\chi_i(L)V_i = 0$ for $i = 1, 2$.

Problem 14 (Fa95, Fa18, Fa24). Show that an $n \times n$ matrix of complex numbers A satisfying

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for $1 \leq i \leq n$ must be invertible.

Problem 15 (Fa14, Fa24). How many zeros does the polynomial $x^{100000} - 1$ have in the finite field $\mathbb{F}_{65537}$? Observe that $65537 = 2^{16} + 1$ is a prime.

Problem 16 (Fa88, Fa97, Fa24). Let the group G be generated by two elements, a and b , both of order 2. Prove that G has a subgroup of index 2.